PROGRAMME CURRICULUM

AI FDE *Fellowship*

Twelve weeks. Six real-world cases. Defended, not just delivered.

12 weeks | 6 cases | 12 assignments | 6 milestone projects | 100 competencies

Sat + Sun, 2 hrs each · Pods of five · One mentor for every two pods
Builder and Engineer tracks · Defence-based grading

THE PROGRAMME

Programme *Overview*

A fellowship for people who deliver AI to real clients, and are graded on defending it.

The Menler AI FDE Fellowship trains Forward Deployed AI Engineers: people who sit with a client, understand a messy operational problem, and own the solution from first interview to handover. You do not learn tools in isolation. You work through six real-world engagements, each built from a public problem in an Indian industry, and you are graded on whether you can defend what you built to a panel that plays the client.

AT A GLANCE

DURATION	12 weeks, Saturday and Sunday, 2 hours each day. 48 live hours in total.
WEEKLY TIME	4 live hours plus 7–8 hours of outside work, about 12 hours a week.
FORMAT	Six two-week case blocks. Each block ends in a Case Defence on the second Sunday.
WORK PRODUCT	12 weekly assignments feed 6 milestone projects, one per case.
PODS	Pods of five learners. One mentor for every two pods.
TRACKS	Builder and Engineer, with identical acceptance criteria.
CURRICULUM	100 competencies, 10 a week, each cleared by an artefact or an oral defence.
GRADING	Defence-based. 80 of 100 competencies plus a capstone pass.
STACK	Claude API, Claude Code, Claude Agent SDK, MCP, n8n, pgvector, Qdrant, Neo4j, promptfoo, Braintrust, Phoenix, Langfuse, OpenTelemetry, Docker, GitHub Actions.

THE SIX CASES

CASE 1	Complaint Triage for a Bank Grievance Cell · BFSI
CASE 2	Vetted-Answer Farm Advisory Copilot · Agriculture
CASE 3	GST Input-Credit Reconciliation Agent · SME finance
CASE 4	Certify a Health Pre-Authorisation Review Assistant · Health insurance
CASE 5	Recover Failed Deliveries for a D2C Brand · E-commerce logistics
CASE 6	Capstone on a fresh brief: provident-fund claims, court judgments, ONDC grievances, or your own client.

THE PARTICIPANT

Who This Fellowship *Is For*

Built for people who will be questioned on their decisions and want to be.

Engineers, analysts, consultants and technical product people who want to deliver AI systems to real clients, and early-career graduates ready to work at that standard. You should be comfortable reading code and working in a terminal, and willing to defend your decisions in front of a panel.

BEFORE WEEK 1

LAPTOP	Able to run Docker, with admin rights.
ACCOUNTS	GitHub and Claude. API credits for the programme are provided by Menler.
DIAGNOSTIC	A short check before the first session confirms that Claude, Claude Code, GitHub and your API key all work.
SKILLS	Python or JavaScript at reading level. Builder-track learners need not have written production code.

TWO TRACKS, ONE STANDARD

BUILDER	You direct Claude Code from starter repositories. You write the plan, review every diff, own the tests and defend the outcome. Suited to analysts, consultants and product people.
ENGINEER	You write the core components yourself and take on stretch items: batching, CI regression, retrievers, connectors, webhook receivers, access controls and instrumentation. Suited to working developers.
SAME BAR	Both tracks meet the same acceptance criteria and face the same Case Defence.

ACCOUNTABILITY: NOBODY HIDES IN THE POD

OWN A COMPONENT	Every learner owns one component of each build and defends it alone at the Case Defence.
ROTATING ROLES	Pod roles rotate each case, so each learner leads client contact, quality and delivery at least once.
CHECKPOINT REPOS	Starting from a Menler checkpoint repository to recover a stalled build caps that case at "Meets".
MENTOR ROLEPLAY	Mentors play client roles by title (head of grievance, extension officer, finance controller). No company is invented.

THE METHOD

How a Case *Block Works*

Two weeks, one project, one defence.

THE TWO-WEEK BLOCK

WEEK A	Frame the problem, learn the concepts the case needs, and build the foundation with your pod. Ends with a project checkpoint.
WEEK B	Complete the build, test it against evidence, harden it, and prepare the defence.
CASE DEFENCE	Second Sunday: 10 minutes walking through the solution, 10 minutes on hidden-scenario results, 10 minutes on a live change request with panel questions.
HIDDEN SCENARIOS	About 10 test scenarios you have never seen per case. They are graded at the defence, and you keep them as a personal Scenario Library.
PROGRESSION	Cases 1–3 are built from scratch. Cases 4–5 hand you a working prototype to make trustworthy and then deployable. Case 6 uses everything on a fresh brief.

THE WEEKLY RHYTHM

SATURDAY · S1	2 live hours. Concepts and build for the first five competencies of the week.
SUNDAY · S2	2 live hours. The next five competencies, with the lab or build that closes the week.
WEEKLY ASSIGNMENT	4–8 hours outside, with your pod. Feeds that week's project milestone.
PROJECT MILESTONE	The slice of the case project due that week. Week A is the foundation, Week B is the finished, defended project.
1:1 BUILD REVIEW	15 minutes with a mentor on your own component.
AI CLINIC	15 minutes to unblock a specific technical problem.
EXECUTIVE SUMMARY	A half-page note for the client each week, so you practise explaining the work to a non-engineer.

THE JOURNEY

12-Week Transformation *Map*

One theme and one identity shift a week. Every week ends with an assignment and a project milestone.

01	**Discover, Frame & Model Basics** *Course taker → Problem framer*	CASE 1
02	**Model Behaviour, Prompting & Context** *Prompt user → Prompt engineer*	CASE 1
03	**Build with Claude Code & Data Foundations** *Prompt engineer → Code director*	CASE 2
04	**Knowledge & Retrieval** *Console builder → Retrieval engineer*	CASE 2
05	**Agent Architecture & Memory** *Retrieval engineer → Agent designer*	CASE 3
06	**Multi-Agent & Enterprise Integration** *Agent designer → Integration engineer*	CASE 3
07	**Evaluation & Quality** *Integration engineer → Evaluator*	CASE 4
08	**Safety, Security & Governance** *Evaluator → Risk owner*	CASE 4
09	**Deploy & Operate** *Risk owner → Operator*	CASE 5
10	**Economics, Delivery & Ownership** *Operator → Owner*	CASE 5
11	**Capstone — Discover & Design** *Owner → Engagement lead*	CASE 6
12	**Capstone — Build, Harden & Demo Day** *Engagement lead → AI FDE*	CASE 6

THE ARC

Frame the problem. Build the system. Prove it works. Defend it to the client.

THE WORK

Assignments & *Projects Index*

Twelve weekly assignments feed six milestone projects. Each project closes with a defence.

WEEK BY WEEK

WEEK 01	4–5 hrs	**Opportunity Memo & Baselines**	Project 1 · Complaint Triage Engine · Foundation checkpoint
WEEK 02	5–6 hrs	**Triage Engine & Benchmark Report**	Project 1 · Complaint Triage Engine · Finished project — Case Defence 1
WEEK 03	6–7 hrs	**Extension-Agent Console & Seeded-Repo Postmortem**	Project 2 · Cited Farm Advisory Copilot · Foundation checkpoint
WEEK 04	6–7 hrs	**Cited Copilot & Bake-off Report**	Project 2 · Cited Farm Advisory Copilot · Finished project — Case Defence 2
WEEK 05	6–7 hrs	**Public MCP Server & Trust-Boundary Diagram**	Project 3 · Reconciliation Agent and Public MCP Server · Foundation checkpoint
WEEK 06	6–7 hrs	**Supervisor System, Review Pack & Integration Tests**	Project 3 · Reconciliation Agent and Public MCP Server · Finished project — Case Defence 3
WEEK 07	6–7 hrs	**Eval Harness, Calibrated Judge & Vendor-Claim Rebuttal**	Project 4 · Certified Review Assistant · Foundation checkpoint
WEEK 08	6–7 hrs	**Hardened Assistant & Go / No-go Certification Memo**	Project 4 · Certified Review Assistant · Finished project — War room and Case Defence 4
WEEK 09	6–7 hrs	**Deployed Pipeline, Dashboards & Pilot Postmortem**	Project 5 · Production NDR Recovery System · Foundation checkpoint
WEEK 10	6–7 hrs	**ROI Model, SOW & Handover Pack**	Project 5 · Production NDR Recovery System · Finished project — Case Defence 5, a CFO panel
WEEK 11	7–8 hrs	**Capstone Design Pack**	Project 6 · Capstone Engagement · Foundation checkpoint
WEEK 12	7–8 hrs	**Capstone Delivery — Ship, Show, Defend**	Project 6 · Capstone Engagement · Finished project — Demo Day

CASE 1 · WEEKS 1–2 · BFSI

Complaint Triage for a Bank *Grievance Cell*

Project 1: Complaint Triage Engine. Built from scratch.

THE REAL-WORLD CONTEXT

RBI Integrated Ombudsman Scheme, FY2024-25: complaints reached about 13.34 lakh, up 13.55% from the year before, and 93.07% were resolved. Banks accounted for 2,41,601 complaints (81.53%) at the Ombudsman offices and NBFCs for 14.80%, with loans and advances the top category and credit cards second. Only 62.16% were maintainable; the rest were duplicates, lacked information, were already dealt with, or fell outside RBI's jurisdiction. A secondary source reports that the RBI Governor said complaints grew about 50% a year over two years while processing capacity grew about 25% (March 2025 remarks). Triage quality is therefore a real bottleneck.

THE CLIENT ASK

A grievance cell (head of customer grievance, compliance officer and contact-centre lead, played by mentors) wants every inbound complaint classified, checked for completeness and duplicates, routed, and given a draft first response, in English, Hindi and Hinglish.

ENGAGEMENT BRIEF

DATA KIT	Synthetic complaints in three language styles, with category mix modelled on the public annual-report distribution, plus a sample of prior complaints and policy snippets.
CONSTRAINTS	The system never closes or rejects a complaint on its own. Every routing decision is explainable. Customer identifiers are masked.
OBJECTIVE	Ship a triage engine that classifies, checks and routes complaints, and prove it beats a classical baseline.
HIDDEN SCENARIOS	Examples of the ~10 you will not see: One complaint carrying two issues · abusive Hinglish · a duplicate of an earlier case · a matter outside the regulator's jurisdiction · missing account details · a customer who pastes an ID number · a message that tries to instruct the triage model.
SUCCESS	Schema validity ⩾99%, per-category F1 with confidence intervals, duplicate and non-maintainable detection reported separately.
CURVEBALL	A new category and Tamil complaints appear mid-block.
BUILDER / ENGINEER	Builder: provided harness and prompt templates. Engineer: adds batching, thinking-budget trade-offs and CI regression.
TIME	8 live + 15 outside hours.

CASE DEFENCE · SECOND SUNDAY

Why did you build it this way, and what happens when it breaks?

10 min solution walk-through · 10 min hidden-scenario results · 10 min live change request and panel Q&A. Focus: Why this pilot first, why this model over the baseline, what you would measure in week one live.

WEEK 1 · DISCOVER

Discover, Frame & *Model Basics*

WEEK OBJECTIVE

Frame Case 1 the way an FDE does: interview the client roles, qualify the candidate uses, set baselines, and learn just enough about how models work to predict where they will fail.

IDENTITY SHIFT *Course taker → Problem framer*

CASE 1 | PROJECT: COMPLAINT TRIAGE ENGINE | WEEK A OF 2

S1 Discover the Problem — Roles, Workflows & Use-Case Qualification

SATURDAY · 2 HRS

BUILDS ON	Setup bridge: Claude, Claude Code, GitHub and your API key are working before the session starts, checked by a short diagnostic.
TOPICS	• FDE operating model and lifecycle • Discovery interviews and process mapping • Use-case qualification (value, feasibility, risk, kill criteria) • AI vs ML vs rules • Success metrics and baselines
CLAUDE IN USE	Claude as research and synthesis partner: summarise the public annual-report data, draft interview guides, and challenge your process map.
BUILD	Two discovery interviews with mentor-played client roles (head of customer grievance, compliance officer) and a first process map of the complaint workflow.
TOOL STACK	Claude · GitHub · a diagramming tool

S2 How Models Work — Tokens, Context & Why They Hallucinate

SUNDAY · 2 HRS

BUILDS ON	S1: you know what the client needs. Now you learn what a model can and cannot do reliably.
TOPICS	• LLM mechanics (tokens, context, sampling) • How models are trained and why they hallucinate • Model landscape and selection • Stakeholder mapping and contradictory asks • Claude as research and synthesis partner
CLAUDE IN USE	Run the same complaint through three models and compare reasoning, output format and failure behaviour.
BUILD	Lab: map where a complaint can be misrouted and what each error costs. Qualify 8 candidate uses, pick the first pilot and one thing to reject. Read back a 200-line repo in plain language.
TOOL STACK	Claude · Claude Code · API console

WEEKLY ASSIGNMENT

TITLE	Opportunity Memo & Baselines
BRIEF	Write the Case 1 Opportunity Memo: the process map, the 8 qualified uses with scores, the pilot you chose, the one use you reject and why, and baselines computed from the public complaint numbers.
DELIVERABLES	• PDF or Doc: memo • Process map • Half-page executive summary for the client
TIME	4–5 hrs
ASSESSED BY	Mentor review in your 15-minute 1:1, checked against the week's ten competencies.
FEEDS INTO	Week 2: the pilot you chose becomes the triage engine you build and benchmark.

PROJECT 1 MILESTONE · FOUNDATION CHECKPOINT

PROJECT	Complaint Triage Engine
DUE THIS WEEK	• Opportunity Memo with the process map and the 8 qualified uses • Chosen pilot, rejected use and written baselines • Project repository created with the synthetic complaint set

WEEK OUTCOME

A framed, defensible pilot with written baselines. You can say why this pilot first, and what you refused to build.

WEEK 2 · PROMPT

Model Behaviour, Prompting *& Context*

WEEK OBJECTIVE

Turn the chosen pilot into a working triage engine and prove it against a classical baseline, using prompts, schemas and context that you control.

IDENTITY SHIFT *Prompt user → Prompt engineer*

CASE 1 | PROJECT: COMPLAINT TRIAGE ENGINE | WEEK B OF 2

S1 Prompts, Schemas & Reasoning — Making Output Dependable

SATURDAY · 2 HRS

BUILDS ON	Week 1: the pilot and its baselines. Now you make the model produce structured, checkable output.
TOPICS	• System prompts and persona design • Structured outputs and schema enforcement • Reasoning models and thinking budgets • Reasoning patterns (CoT, ReAct, Reflexion) • Context budgeting and compaction
CLAUDE IN USE	Claude with JSON-schema output returning category, issue count, maintainability reason, language, missing information, urgency, suggested team and confidence.
BUILD	Triage engine v0 returning schema-valid JSON on a golden set of synthetic complaints in English, Hindi and Hinglish.
TOOL STACK	Claude API · JSON Schema · Claude Code · Git

S2 Cost, Grounding & Benchmarks — Beating the Baseline

SUNDAY · 2 HRS

BUILDS ON	S1: the engine works. Now you measure it honestly against what the client could do without AI.
TOPICS	• Prompt caching and batching • Grounding and citations • Prompt versioning and regression • Multilingual and Indic model choices • LLM vs classical-baseline benchmark
CLAUDE IN USE	Claude as labeller and error analyst. You verify every label it proposes.
BUILD	Benchmark three models, a TF-IDF baseline and rules on the golden set; report cost per 1,000 complaints and p95 latency. Includes a 90-minute timed take-home.
TOOL STACK	Claude API · a TF-IDF baseline · Git · a regression test

WEEKLY ASSIGNMENT

TITLE	Triage Engine & Benchmark Report
BRIEF	Ship triage engine v0 in a prompt repo with a regression test. Report per-category F1 with confidence intervals, cost per 1,000 complaints and p95 latency, and write the top-10 failure catalogue.
DELIVERABLES	• Repo link • Benchmark report • Half-page executive summary for the client
TIME	5–6 hrs
ASSESSED BY	Mentor review in your 15-minute 1:1, checked against the week's ten competencies.
FEEDS INTO	Case Defence 1 on Sunday. Week 3 reuses your repo discipline for Case 2.

PROJECT 1 MILESTONE · FINISHED PROJECT

PROJECT	Complaint Triage Engine
DUE THIS WEEK	• Triage engine v0 returning schema-valid JSON • Benchmark report: three models, a TF-IDF baseline and rules • Prompt repo with a regression test • Top-10 failure catalogue
GATE	Case Defence 1

COMPONENT OWNER	The component you own and defend alone: _____
POD ROLES	Client lead · Quality lead · Delivery lead (rotates each case)
MENTOR SIGN-OFF	Rubric level: Not yet / Meets / Strong / Hire-ready Date: _____
PORTFOLIO	Repository, report and defence record added to your public portfolio.

WEEK OUTCOME

A benchmarked engine and a failure catalogue that tells you exactly what to fix next.

CASE 2 · WEEKS 3–4 · AGRICULTURE

Vetted-Answer Farm *Advisory Copilot*

Project 2: Cited Farm Advisory Copilot. Built from scratch.

THE REAL-WORLD CONTEXT

Digital Green and Gooey.AI built Farmer.Chat, a GPT-4-based assistant trained only on trusted, local resources (farmer videos, government extension documents, call-centre transcripts and manuals). It accepts text, audio and images in languages including Hindi, Telugu and Bhojpuri, and was piloted with extension workers before being scaled to farmers. OpenAI's customer story gives early estimates that it could cut extension cost from $35 to $0.35 per farmer; that is a claim, not a measured result. Wadhwani AI has also explored generative AI for Kisan call centres.

THE CLIENT ASK

An agricultural extension programme (programme head and a state extension officer, played by mentors) wants a copilot, first for extension agents and later for farmers. It must answer only from vetted package-of-practices documents, in local languages, with sources, and say "I don't know" when the material does not cover a question.

ENGAGEMENT BRIEF

DATA KIT	Public-style crop advisories in PDF (some scanned, some conflicting between seasons), synthetic call-centre transcripts, a small video-transcript set, and synthetic farmer profiles.
CONSTRAINTS	Pesticide and dosage questions must cite a source and point to a human officer. Farmer phone numbers and IDs are protected (DPDP basics). The system must be usable by agents with low-end phones.
OBJECTIVE	Ship a console and a copilot that answers only from vetted sources, cites every claim and abstains correctly.
HIDDEN SCENARIOS	Examples of the ~10 you will not see: A Hindi query with spelling variants · two advisories that disagree by season · a question the corpus cannot answer · a dosage question · a farmer who shares an ID number · a stale advisory after an update.
SUCCESS	Faithfulness ⊠90% on answerable questions, ⊠85% correct abstention, every claim cited.
CURVEBALL	A new advisory replaces an old one and stale answers must disappear within an hour.
BUILDER / ENGINEER	Builder: provided pipeline skeleton. Engineer: writes the retriever and reranker and adds a knowledge-graph path.
TIME	8 live + 16 outside hours.

CASE DEFENCE · SECOND SUNDAY

Why did you build it this way, and what happens when it breaks?

10 min solution walk-through · 10 min hidden-scenario results · 10 min live change request and panel Q&A. Focus: Why your chunking and embedding choices, and show one retrieval failure and your fix.

Build with Claude Code & *Data Foundations*

WEEK OBJECTIVE

Learn to direct Claude Code like a lead engineer — plan, implement, test and review — and stand up a full-stack console for the farm advisory case on a clean data foundation.

IDENTITY SHIFT *Prompt engineer → Code director*

CASE 2 | PROJECT: CITED FARM ADVISORY COPILOT | WEEK A OF 2

S1 Claude Code Workflow — Plan, Implement, Test, Review

SATURDAY · 2 HRS

BUILDS ON	Weeks 1–2: you framed a problem and benchmarked an engine. Now you build full applications around the model.
TOPICS	• Claude Code workflow (CLAUDE.md, plan, implement, test) • Extending Claude Code (skills, subagents, hooks, plugins) • Full-stack app (UI, API, auth) • Git and diff review of AI-written code • Testing (unit, integration, test-first with AI)
CLAUDE IN USE	Claude Code as implementer. You write the plan and review every diff before it merges.
BUILD	Break-it lab: a seeded repo with 5 bugs, a leaked key and one hallucinated dependency. Find them, fix them and write a postmortem.
TOOL STACK	Claude Code · Git · GitHub

S2 Data Foundations — Schemas, Ingestion, PII & UX

SUNDAY · 2 HRS

BUILDS ON	S1: you can direct and review code. Now the data underneath it has to be trustworthy and protected.
TOPICS	• SQL, schema design, data modelling • Data quality and ingestion • PII classification and DPDP basics • AI UX (confidence, fallbacks, feedback) • PRD-lite and SOW-lite
CLAUDE IN USE	Claude drafts schemas, validators and the PRD-lite. You decide what counts as PII.
BUILD	The extension-agent console: login, document library, question screen, feedback buttons, a Postgres schema with PII tagged, validated ingestion, CI and a deployed URL.
TOOL STACK	Postgres · Claude Code · CI · a hosting service

WEEKLY ASSIGNMENT

TITLE	Extension-Agent Console & Seeded-Repo Postmortem
BRIEF	Deploy the console with a PII-tagged schema and validated ingestion. Write the postmortem of the seeded-repo lab: the 5 bugs, the leaked key and the hallucinated dependency.
DELIVERABLES	• Deployed URL • Repo link • Postmortem • Half-page executive summary for the client
TIME	6–7 hrs
ASSESSED BY	Mentor review in your 15-minute 1:1, checked against the week's ten competencies.
FEEDS INTO	Week 4 adds the retrieval layer and the abstaining copilot inside this console.

PROJECT 2 MILESTONE · FOUNDATION CHECKPOINT

PROJECT	Cited Farm Advisory Copilot
DUE THIS WEEK	• Extension-agent console deployed, with login and document library • PII-tagged Postgres schema and validated ingestion • Seeded-repo postmortem

WEEK OUTCOME

A deployed console you can extend, and proof that you catch what AI-written code gets wrong.

WEEK 4 · RETRIEVE

Knowledge & *Retrieval*

WEEK OBJECTIVE

Give the console a knowledge layer that answers only from vetted documents, cites every claim and says "I don't know" when the corpus is silent.

IDENTITY SHIFT *Console builder → Retrieval engineer*

CASE 2 | PROJECT: CITED FARM ADVISORY COPILOT | WEEK B OF 2

S1 Ingest & Index — Parsing, Chunking & Embeddings

SATURDAY · 2 HRS

BUILDS ON	Week 3: the console and a clean schema. Now the messy advisories have to become searchable knowledge.
TOPICS	• Parsing and ingestion (PDF, OCR, tables) • Chunking • Embeddings and model choice • Vector stores and ANN • Freshness, dedup and conflicting sources
CLAUDE IN USE	Claude extracts tables and normalises scanned advisories. You sample-check the extraction.
BUILD	Ingest the messy advisories (some scanned, some conflicting between seasons) into a vector store with source and date metadata.
TOOL STACK	pgvector · Qdrant · an OCR tool · Claude

S2 Retrieve, Cite & Abstain — Hybrid Search, Reranking & Failur

SUNDAY · 2 HRS

BUILDS ON	S1: the knowledge is indexed. Now answers must be grounded, cited and refused when unsupported.
TOPICS	• Hybrid search and reranking • Citations and grounded answers • Retrieval metrics and failure analysis • Query rewriting and agentic RAG • Knowledge graphs
CLAUDE IN USE	Claude generates answers constrained to retrieved passages and refuses when evidence is absent.
BUILD	Retrieval bake-off (3 chunking strategies × 2 embeddings × reranking on/off, 100 questions), then a 60-minute timed sprint on a new crop's documents. Ship the cited, abstaining copilot inside the console.
TOOL STACK	pgvector · Qdrant · a reranker · Neo4j (optional graph path)

WEEKLY ASSIGNMENT

TITLE	Cited Copilot & Bake-off Report
BRIEF	Deliver the cited, abstaining copilot, the bake-off report, a PRD-lite and SOW-lite, and a rollout plan that puts extension agents first and farmers second.
DELIVERABLES	• Deployed copilot • Repo • Report • Half-page executive summary for the client
TIME	6–7 hrs
ASSESSED BY	Mentor review in your 15-minute 1:1, checked against the week's ten competencies.
FEEDS INTO	Case Defence 2 on Sunday. Week 5 turns retrieval into one tool among several that an agent can call.

PROJECT 2 MILESTONE · FINISHED PROJECT

PROJECT	Cited Farm Advisory Copilot
DUE THIS WEEK	• Cited, abstaining copilot running inside the console • Retrieval bake-off report on 100 questions • PRD-lite and SOW-lite • Rollout plan: extension agents first, farmers second
GATE	Case Defence 2

COMPONENT OWNER	The component you own and defend alone: _____
POD ROLES	Client lead · Quality lead · Delivery lead (rotates each case)
MENTOR SIGN-OFF	Rubric level: Not yet / Meets / Strong / Hire-ready Date: _____
PORTFOLIO	Repository, report and defence record added to your public portfolio.

WEEK OUTCOME

A copilot that tells you what it knows, where it learned it and when it does not know.

CASE 3 · WEEKS 5–6 · SME FINANCE

GST Input-Credit *Reconciliation Agent*

Project 3: Reconciliation Agent and Public MCP Server. Built from scratch.

THE REAL-WORLD CONTEXT

GSTR-2B is a static monthly statement. Mismatches commonly arise when suppliers file GSTR-1 late (pushing credit to the next cycle), report B2B invoices as B2C, or lack a valid IRN, and the fix is to ask suppliers to amend. Reconciliation is best done monthly, before filing GSTR-3B, and a supplier's credit note reduces the available credit. Credit that goes unreconciled can be lost permanently, so this is a recurring chase-and-verify workload for SME finance teams. GST rules change often, so a chartered accountant reviews this brief before each cohort.

THE CLIENT ASK

A finance controller and a chartered accountant (played by mentors) want an agent that reconciles the purchase register against GSTR-2B each month, explains every mismatch, chases suppliers and prepares a review pack. It never files returns and never decides credit eligibility. A human approves every credit decision.

ENGAGEMENT BRIEF

DATA KIT	Synthetic purchase register (spreadsheet export), GSTR-2B-style JSON, a supplier master with contacts, a sandbox ERP, and a WhatsApp and email sandbox.
CONSTRAINTS	No credit decision without human approval. No return is ever filed by the agent. Every supplier message is logged and sent at most once.
OBJECTIVE	Ship a reconciliation agent behind a published MCP server, with approval gates and safe integrations.
HIDDEN SCENARIOS	Examples of the ~10 you will not see: A GSTIN typo · an invoice that lands in next month's statement · a credit note · a duplicate register line · a supplier who says "already paid" with no proof · a request to "just claim it" · a duplicate webhook.
SUCCESS	Zero credit decisions without human approval, zero double-sent supplier messages, mismatch explanations reviewed by the pod's "accountant".
CURVEBALL	The ERP goes down on filing day.
BUILDER / ENGINEER	Builder: agent scaffold and n8n templates. Engineer: writes the webhook receiver, queue and a connector.
TIME	8 live + 15 outside hours. Gate: the mid-point defence closes this block.

CASE DEFENCE · SECOND SUNDAY

Why did you build it this way, and what happens when it breaks?

10 min solution walk-through · 10 min hidden-scenario results · 10 min live change request and panel Q&A. Focus: Why this autonomy boundary, and what happens when a supplier lies.

WEEK 5 · AGENTS

Agent Architecture *& Memory*

WEEK OBJECTIVE

Design an agent with typed tools, bounded autonomy and human approvals, and publish its tools as an MCP server.

IDENTITY SHIFT *Retrieval engineer → Agent designer*

CASE 3 | PROJECT: RECONCILIATION AGENT AND PUBLIC MCP SERVER | WEEK A OF 2

S1 Agent Architecture — Tools, Loops & Autonomy Boundaries

SATURDAY · 2 HRS

BUILDS ON	Week 4: retrieval as a single capability. Now you let a model take actions, within limits you set.
TOPICS	• Tool design and schemas • Agent loop and bounded autonomy • Workflow vs agent decision • Human-in-the-loop approvals • Failure handling
CLAUDE IN USE	Claude Agent SDK and tool use: the model proposes, deterministic code executes, a human approves.
BUILD	Typed tools (load_register, load_2b, match_invoices, explain_mismatch, draft_supplier_message, log_decision), with rules-first matching and the LLM reserved for exceptions.
TOOL STACK	Claude Agent SDK · Claude API tool use · Git

S2 MCP & Memory — Publishing Your Tools

SUNDAY · 2 HRS

BUILDS ON	S1: the tools exist. Now they become a reusable, permissioned server and the agent gets memory.
TOPICS	• Building an MCP server (2026-07-28 stateless spec) • MCP authorization and tool permissions • Short-term and long-term memory • Durable workflows • Computer-use and browser agents for systems without APIs
CLAUDE IN USE	Claude Code scaffolds the server. You write the permission model and the approval gates.
BUILD	Publish the MCP server on GitHub with approval gates, a kill switch and a trust-boundary diagram with SLAs.
TOOL STACK	MCP · Claude Code · GitHub

WEEKLY ASSIGNMENT

TITLE	Public MCP Server & Trust-Boundary Diagram
BRIEF	Publish the reconciliation tools as an MCP server with typed schemas, approval gates and a kill switch, and draw the trust-boundary diagram with the SLAs each boundary carries.
DELIVERABLES	• GitHub link • Diagram • Half-page executive summary for the client
TIME	6–7 hrs
ASSESSED BY	Mentor review in your 15-minute 1:1, checked against the week's ten competencies.
FEEDS INTO	Week 6 adds a supervisor, a supplier-comms agent and a reviewer, and connects them to enterprise systems.

PROJECT 3 MILESTONE · FOUNDATION CHECKPOINT

PROJECT	Reconciliation Agent and Public MCP Server
DUE THIS WEEK	• Typed reconciliation tools with rules-first matching • MCP server published on GitHub • Approval gates, kill switch and trust-boundary diagram

WEEK OUTCOME

A public MCP server and a clear answer to where the agent stops and a person starts.

WEEK 6 · INTEGRATE

Multi-Agent & *Enterprise Integration*

WEEK OBJECTIVE

Orchestrate several agents against messy enterprise systems, and make every integration idempotent, observable and safe to retry.

IDENTITY SHIFT *Agent designer → Integration engineer*

CASE 3 | PROJECT: RECONCILIATION AGENT AND PUBLIC MCP SERVER | WEEK B OF 2

S1 Multi-Agent Patterns — Supervisor, Sub-agents & Shared State

SATURDAY · 2 HRS

BUILDS ON	Week 5: one agent with typed tools. Now several agents with different jobs.
TOPICS	• Multi-agent patterns (router, supervisor, parallel, handoff) • Subagents and the Claude Agent SDK • Shared state and communication • Channels (WhatsApp, email) • Integration testing with sandboxes
CLAUDE IN USE	Claude Agent SDK subagents: a matcher, a supplier-comms agent and a reviewer coordinated by a supervisor.
BUILD	A supervisor with a matcher, a supplier-comms agent and a reviewer, tested against the sandbox ERP and the WhatsApp and email sandboxes.
TOOL STACK	Claude Agent SDK · MCP · sandboxes

S2 Enterprise Integration — Webhooks, Queues & Legacy Systems

SUNDAY · 2 HRS

BUILDS ON	S1: the agents coordinate. Now the outside world retries, duplicates and fails.
TOPICS	• REST/GraphQL, webhooks, idempotency • OAuth2 and connectors • Workflow orchestration and queues (n8n) • Legacy systems (SOAP/XML, SQL Server) • Rate limits, backoff, dead-letter handling
CLAUDE IN USE	Claude reads an unfamiliar codebase and drafts the audit, but you own the fix plan.
BUILD	Lab: inherit an undocumented reconciliation script, audit it and write a safe fix plan first. Then build an idempotent webhook receiver and queue for supplier replies, an ERP lookup and a human handoff pack.
TOOL STACK	n8n · REST · OAuth2 · sandbox ERP · queue

WEEKLY ASSIGNMENT

TITLE	Supervisor System, Review Pack & Integration Tests
BRIEF	Deliver the supervisor system, the review pack template, the integration test pack and the runbook. Show that a duplicate webhook cannot double-send a supplier message.
DELIVERABLES	• Repo • Test pack • Runbook • Half-page executive summary for the client
TIME	6–7 hrs
ASSESSED BY	Mentor review in your 15-minute 1:1, checked against the week's ten competencies.
FEEDS INTO	Mid-point defence on Sunday. Week 7 moves from building to judging whether an AI system is good enough.

PROJECT 3 MILESTONE · FINISHED PROJECT

PROJECT	Reconciliation Agent and Public MCP Server
DUE THIS WEEK	• Supervisor system with matcher, supplier-comms agent and reviewer • Review pack template and integration test pack • Runbook • Evidence that a duplicate webhook cannot double-send
GATE	Mid-point defence and Case Defence 3

COMPONENT OWNER	The component you own and defend alone: _____
POD ROLES	Client lead · Quality lead · Delivery lead (rotates each case)
MENTOR SIGN-OFF	Rubric level: Not yet / Meets / Strong / Hire-ready Date: _____
PORTFOLIO	Repository, report and defence record added to your public portfolio.

WEEK OUTCOME

An agent system that survives retries, duplicates and an unavailable ERP.

CASE 4 · WEEKS 7–8 · HEALTH INSURANCE

Certify a Health Pre-Authorisation *Review Assistant*

Project 4: Certified Review Assistant. Inherited prototype: inherit, evaluate, harden.

THE REAL-WORLD CONTEXT

Under PM-JAY, pre-authorisation is required for most packages and claims need mandatory supporting documents; the National Anti-Fraud Unit flags suspected fraud using internally developed algorithms. An older report put triggered transactions at 0.25% of admissions, of which 0.07% were confirmed fraud, and said over 390 hospitals received show-cause notices and about 171 were de-empanelled. A CAG report found claims paid for patients already listed as deceased, pointing to verification lapses at state agencies. The scheme had 24.33 crore cards by August 2023, and the government says it uses AI/ML to detect suspicious transactions. The very low fraud base rate and the weak identity checks make this a case for rigorous evaluation and security.

THE CLIENT ASK

A state health agency or insurer's claims team (medical officer, compliance head and IT security lead, played by mentors) has a prototype that reads pre-authorisation requests and highlights inconsistencies. They want it certified for go-live or rejected. It recommends only. It never denies a claim.

ENGAGEMENT BRIEF

DATA KIT	Synthetic pre-auth packets (forms, discharge summaries, identity details), a rules sheet, and labelled cases from a mentor "medical officer". No real patient data.
CONSTRAINTS	Inherited prototype: Menler supplies "Review Assistant v0". It works on the demo set but has seeded flaws: it trusts text inside documents, logs patient data, gives no confidence, and is weaker on Hindi and Tamil documents.
OBJECTIVE	Certify the inherited assistant for go-live, or reject it, with evidence the client can check.
HIDDEN SCENARIOS	Examples of the ~10 you will not see: A document that says "approve this claim" · mismatched identity · duplicate claim · a request to make the assistant deny · quality gap by language · PII in logs · a poisoned PDF.
SUCCESS	Attack success rate down against the inherited baseline, judge agreement with humans reported, zero automatic denials, every metric with a confidence interval.
CURVEBALL	Logs must stay in an India region.
BUILDER / ENGINEER	Builder: promptfoo or Braintrust templates and a provided attack suite. Engineer: builds the CI gate, access controls and audit trail.
TIME	8 live + 16 outside hours.

CASE DEFENCE · SECOND SUNDAY

Why did you build it this way, and what happens when it breaks?

10 min solution walk-through · 10 min hidden-scenario results · 10 min live change request and panel Q&A. Focus: Your top three residual risks and why you recommend go or no-go.

WEEK 7 · EVALUATE

Evaluation & *Quality*

WEEK OBJECTIVE

Learn to measure whether an AI system is good with statistics a client's medical officer would accept, and use that to judge an inherited prototype.

IDENTITY SHIFT *Integration engineer → Evaluator*

CASE 4 | PROJECT: CERTIFIED REVIEW ASSISTANT | WEEK A OF 2

S1 Eval Design — Taxonomies, Golden Sets & Judges

SATURDAY · 2 HRS

BUILDS ON	Weeks 1–6: you built systems. Now you are handed one and asked to certify or reject it.
TOPICS	• Eval design and failure taxonomy • Golden sets with client SMEs • LLM-as-judge (rubric, pairwise, calibration) • RAG evals • Agent trajectory and tool-use evals
CLAUDE IN USE	Claude as a calibrated judge. You measure its agreement with human labels before you trust it.
BUILD	Label 150 cases with a pod partner against the mentor medical officer's rules sheet and report inter-annotator agreement.
TOOL STACK	promptfoo · Braintrust · Claude (judge) · Git

S2 Eval Statistics & Gates — Low-Prevalence Reality

SUNDAY · 2 HRS

BUILDS ON	S1: you have labels and a judge. Now the numbers have to survive scrutiny.
TOPICS	• Eval statistics (sample size, variance, annotator agreement) • CI regression gates • Online evals, feedback and drift • Eval platforms and experiment tracking • Reporting quality to a client
CLAUDE IN USE	Claude drafts the quality report. You check every figure and every interval.
BUILD	Eval harness for a low-prevalence problem (precision at a 0.07% base rate, review workload), sliced by language, hospital type and package. Audit a vendor's "95% accuracy" claim and write the rebuttal.
TOOL STACK	promptfoo · Braintrust · RAGAS · CI

WEEKLY ASSIGNMENT

TITLE	Eval Harness, Calibrated Judge & Vendor-Claim Rebuttal
BRIEF	Deliver the eval harness, the calibrated judge with its human-agreement figure, a quality report with confidence intervals on every metric, and the written rebuttal of the vendor's accuracy claim.
DELIVERABLES	• Repo • Report • Rebuttal • Half-page executive summary for the client
TIME	6–7 hrs
ASSESSED BY	Mentor review in your 15-minute 1:1, checked against the week's ten competencies.
FEEDS INTO	Week 8: the failures you found become the attack surface you harden.

PROJECT 4 MILESTONE · FOUNDATION CHECKPOINT

PROJECT	Certified Review Assistant
DUE THIS WEEK	• 150 labelled cases and annotator agreement • Eval harness for the low-prevalence problem • Calibrated judge with its human-agreement figure • Vendor-claim rebuttal

WEEK OUTCOME

A defensible quality verdict, with the uncertainty stated, for a problem where accuracy alone is meaningless.

WEEK 8 · SECURE

Safety, Security *& Governance*

WEEK OBJECTIVE

Threat-model the inherited assistant, attack it, harden it, and decide with evidence whether it should go live.

IDENTITY SHIFT *Evaluator → Risk owner*

CASE 4 | PROJECT: CERTIFIED REVIEW ASSISTANT | WEEK B OF 2

S1 Threats & Attacks — Injection, Tool Abuse & the War Room

SATURDAY · 2 HRS

BUILDS ON	Week 7: you know how good the prototype is. Now you find out how it breaks under attack.
TOPICS	• Threat modelling (OWASP LLM Top 10) • Prompt injection, direct and indirect • Red-team method and war room • Tool abuse, least privilege, blast radius • Bias, fairness, abuse testing
CLAUDE IN USE	Claude as attacker and as defender, in separate sessions with separate instructions.
BUILD	A 15-threat model and 40 attacks, including instructions hidden in a discharge summary, an identity mismatch and duplicate claims across hospitals.
TOOL STACK	Attack suite · promptfoo · Claude

S2 Controls & Governance — Guardrails, Access & DPDP

SUNDAY · 2 HRS

BUILDS ON	S1: you have a ranked list of failures. Now you add controls and write the decision.
TOPICS	• Guardrails and PII redaction • Identity and permissions (RBAC, permission-aware retrieval) • Data residency, retention, DPDP and sector rules • Audit trails, lineage, explainability • Risk review and security questionnaire
CLAUDE IN USE	Claude drafts the DPDP impact assessment and security questionnaire. You verify against the actual system. DPDP Rules were notified in November 2025, with obligations phasing in over 18 months to about May 2027; breach reporting within 72 hours is among them.
BUILD	Guardrails, PII-free logs, role-based access, an immutable audit trail and a DPDP impact assessment. War room: pods attack each other's hardened versions.
TOOL STACK	Presidio · NeMo Guardrails · RBAC · audit log

WEEKLY ASSIGNMENT

TITLE	Hardened Assistant & Go / No-go Certification Memo
BRIEF	Deliver the hardened assistant, the threat model, the risk register, the completed security questionnaire and a go / no-go certification memo for the client naming your top three residual risks.
DELIVERABLES	• Repo • Certification memo (PDF) • Half-page executive summary for the client
TIME	6–7 hrs
ASSESSED BY	Mentor review in your 15-minute 1:1, checked against the week's ten competencies.
FEEDS INTO	Case Defence 4 on Sunday. Week 9 takes a different inherited prototype from a laptop to production.

PROJECT 4 MILESTONE · FINISHED PROJECT

PROJECT	Certified Review Assistant
DUE THIS WEEK	• Hardened assistant with guardrails, RBAC and audit trail • 15-threat model, 40 attacks and risk register • Security questionnaire and DPDP impact assessment • Go / no-go certification memo
GATE	War room and Case Defence 4

COMPONENT OWNER	The component you own and defend alone: _____
POD ROLES	Client lead · Quality lead · Delivery lead (rotates each case)
MENTOR SIGN-OFF	Rubric level: Not yet / Meets / Strong / Hire-ready Date: _____
PORTFOLIO	Repository, report and defence record added to your public portfolio.

WEEK OUTCOME

A signed recommendation you can defend, including when the honest answer is no-go.

CASE 5 · WEEKS 9–10 · E-COMMERCE LOGISTICS

Recover Failed Deliveries for a *D2C Brand*

Project 5: Production NDR Recovery System. Inherited prototype: inherit, deploy, prove roi.

THE REAL-WORLD CONTEXT

Roughly 60–65% of Indian e-commerce orders are cash on delivery. A failed delivery raises an NDR (non-delivery report), and a common practice is a 24-hour response SLA with escalation after the second failed attempt. Typical RTO rates in India run about 20–30%, though they vary by category, region and payment mix. A logistics vendor claims automated NDR workflows can recover 30–40% of failed deliveries. That is a vendor claim, so learners must measure real lift.

THE CLIENT ASK

A D2C brand's logistics and CX head and its finance lead (played by mentors) want an outreach agent. On an NDR event it contacts the customer on WhatsApp or voice to confirm address, time and COD intent, tells the courier to reattempt, and hands off to a human when needed. They want it live, reliable and proven worth its cost.

ENGAGEMENT BRIEF

DATA KIT	A synthetic NDR event generator (reason codes, COD or prepaid, pincodes, attempt counts), a courier webhook simulator, a WhatsApp sandbox and per-message costs.
CONSTRAINTS	Inherited prototype: Menler supplies "NDR Outreach v0". It works on a laptop with no monitoring, no cost limits, no opt-out handling and one hard-coded message. WhatsApp opt-in, template approval and opt-out handling apply, and the live pilot needs a consent notice.
OBJECTIVE	Take the inherited prototype to production and prove incremental recovery and ROI.
HIDDEN SCENARIOS	Examples of the ~10 you will not see: A customer replies "STOP" · "call me tomorrow" at 11 pm · a cancellation request mid-NDR · an address change after the courier cut-off · template rejection or rate limiting · a 10× festive-season spike · duplicate courier webhooks.
SUCCESS	p95 latency within SLO, recovery lift against holdout with a confidence interval, cost per recovered shipment under the margin at risk, opt-outs honoured immediately.
CURVEBALL	The client's budget is cut by 60%.
BUILDER / ENGINEER	Builder: deploy template and managed hosting. Engineer: writes the pipeline and OpenTelemetry instrumentation.
TIME	8 live + 16 outside hours.

CASE DEFENCE · SECOND SUNDAY

Why did you build it this way, and what happens when it breaks?

10 min solution walk-through · 10 min hidden-scenario results · 10 min live change request and panel Q&A. Focus: A CFO panel on every number in your ROI model, and a recorded handover another pod can follow.

WEEK 9 · OPERATE

Deploy & *Operate*

WEEK OBJECTIVE

Take an inherited prototype from a laptop to a monitored, cost-capped, rollback-ready service, and run it against real conversations.

IDENTITY SHIFT *Risk owner → Operator*

CASE 5 | PROJECT: PRODUCTION NDR RECOVERY SYSTEM | WEEK A OF 2

S1 Ship It — Containers, CI/CD & Cloud

SATURDAY · 2 HRS

BUILDS ON	Weeks 7–8: you judged and hardened. Now you make something runnable by someone else, safely.
TOPICS	• Environments, secrets, config • Docker • CI/CD • Cloud deployment with IAM and networking basics • Release safety (canary, A/B, shadow, rollback)
CLAUDE IN USE	Claude Code writes the Dockerfile and pipeline. You decide the gates and own the rollback.
BUILD	Containerise NDR Outreach v0, add CI/CD with eval and security gates, canary a new message variant and roll back in under 5 minutes.
TOOL STACK	Docker · GitHub Actions · a cloud host

S2 Run It — Observability, SLOs, Cost & Incidents

SUNDAY · 2 HRS

BUILDS ON	S1: the system is deployed. Now you prove you can see it, afford it and fix it.
TOPICS	• Observability (logs, traces, token and latency metrics) • SLOs, error budgets, alerting • Cost control (caching, batching, routing, budget guards) • Incident response and postmortems • UAT plan, runbook, operational handoff
CLAUDE IN USE	Claude summarises traces during the incident drill. You make the call.
BUILD	Dashboards (latency, cost per event, response rate), the 24-hour NDR SLO, budget caps and a kill switch. Live incident drill: provider outage plus a cost spike. Live pilot: one week on a real design-partner store if available; otherwise Menler's own consented WhatsApp follow-up flows run the same event → message → outcome pipeline.
TOOL STACK	OpenTelemetry · Phoenix · Langfuse · WhatsApp sandbox

WEEKLY ASSIGNMENT

TITLE	Deployed Pipeline, Dashboards & Pilot Postmortem
BRIEF	Deliver the deployed pipeline with CI gates, the latency / cost / response dashboards, the budget caps and kill switch, and a postmortem covering the incident drill and the live pilot.
DELIVERABLES	• Deployed URL • Dashboards • Postmortem • Half-page executive summary for the client
TIME	6–7 hrs
ASSESSED BY	Mentor review in your 15-minute 1:1, checked against the week's ten competencies.
FEEDS INTO	Week 10 turns the pilot data into an ROI argument and a handover.

PROJECT 5 MILESTONE · FOUNDATION CHECKPOINT

PROJECT	Production NDR Recovery System
DUE THIS WEEK	• Containerised pipeline with CI/CD, eval and security gates • Latency, cost and response dashboards • Budget caps and kill switch • Incident drill and live pilot notes

WEEK OUTCOME

A production system with an SLO, a budget, a rollback and a postmortem.

WEEK 10 · OWN

Economics, Delivery *& Ownership*

WEEK OBJECTIVE

Prove that the system pays for itself, negotiate the commercial terms, and hand it over so that another team can run it.

IDENTITY SHIFT *Operator → Owner*

CASE 5 | PROJECT: PRODUCTION NDR RECOVERY SYSTEM | WEEK B OF 2

S1 Economics & Evidence — Unit Costs, Routing & ROI

SATURDAY · 2 HRS

BUILDS ON	Week 9: a running pilot with data. Now you turn data into a business case.
TOPICS	• Model selection matrix and routing • Unit economics per request • ROI proof and business case • Vendor and model risk, and the fine-tune vs RAG vs prompt decision • Adoption metrics and change management
CLAUDE IN USE	Claude as spreadsheet and sensitivity-analysis partner. You own every assumption.
BUILD	An A/B test with a holdout to measure incremental recovery rather than reply rate; an ROI model (cost per recovered shipment vs forward and return shipping cost; best, base, worst cases); a routing recommendation.
TOOL STACK	A spreadsheet · Claude · experiment notebook

S2 Delivery & Ownership — SOW, Negotiation & Handover

SUNDAY · 2 HRS

BUILDS ON	S1: the numbers are defensible. Now you commit to terms and make the system someone else's.
TOPICS	• SOW, commercials, pricing negotiation • Executive readout • Handover and enablement • Account expansion • Legal, IP and licensing awareness
CLAUDE IN USE	Claude drafts the SOW and training script. You negotiate and you present.
BUILD	A phase-2 SOW, a handover pack (runbook, admin guide, recorded 20-minute training), a scored roadmap of 3 next uses, and a negotiation of scope and price with a real SMB owner.
TOOL STACK	Claude · screen recorder · docs

WEEKLY ASSIGNMENT

TITLE	ROI Model, SOW & Handover Pack
BRIEF	Deliver the ROI model, the phase-2 SOW, the executive readout, the handover pack with the recorded training, and one written case study of the engagement.
DELIVERABLES	• Pack of documents • Recording • Half-page executive summary for the client
TIME	6–7 hrs
ASSESSED BY	Mentor review in your 15-minute 1:1, checked against the week's ten competencies.
FEEDS INTO	Case Defence 5 on Sunday (a CFO panel). Week 11 starts the capstone on a fresh brief.

PROJECT 5 MILESTONE · FINISHED PROJECT

PROJECT	Production NDR Recovery System
DUE THIS WEEK	• A/B test with holdout and ROI model • Phase-2 SOW and executive readout • Handover pack with recorded 20-minute training • Written case study
GATE	Case Defence 5, a CFO panel

COMPONENT OWNER	The component you own and defend alone: _____
POD ROLES	Client lead · Quality lead · Delivery lead (rotates each case)
MENTOR SIGN-OFF	Rubric level: Not yet / Meets / Strong / Hire-ready Date: _____
PORTFOLIO	Repository, report and defence record added to your public portfolio.

WEEK OUTCOME

A system a client can run, a number a CFO can check, and a contract you negotiated yourself.

CASE 6 · WEEKS 11–12 · CAPSTONE

Capstone: A *Fresh Brief*

Project 6: Capstone Engagement.

HOW THE CAPSTONE WORKS

Pods pick one brief and run the full lifecycle: Discover, Design, Build, Harden, Deliver. There is no starter prototype. The pod agrees scope with a client role inside a 48-hour discovery window, then builds against hidden scenarios.

THE FOUR BRIEFS

BRIEF A — Member claim-readiness assistant for a provident-fund body

EPFO processed over 5.08 crore claims in FY2024-25, with auto-settled claims roughly doubling to 1.87 crore. Rejection reasons were cut from 44 to 18, and the auto-settlement limit for several advance types rose to Rs 5 lakh. Settlement time fell from about 10 days to 3–4 days, and the main complaint themes were changes to personal details and fund withdrawals. Build: a checker that tells a member why a claim is likely to be rejected (name, date-of-birth, KYC mismatches) and what to fix, plus an exception-triage queue for staff. It explains and routes but never approves or rejects.

BRIEF B — Judgment research assistant for district-court users

As of 31 December 2025, 4,76,57,328 cases were pending in district and subordinate courts and 63,66,023 in High Courts. SUVAS is an AI tool that translates Supreme Court judgments into regional languages. Build: a retrieval and summary assistant over a public judgments corpus with verified citations only: every case name and quote must exist in the corpus, otherwise the assistant says so. It is a research aid, not legal advice. Check the licence terms of any corpus first.

BRIEF C — Seller-side grievance copilot for the ONDC network

ONDC's Issue and Grievance Management framework went live on 1 September 2023 with online dispute resolution. It has three stages: internal resolution, the network participant's grievance redress officer, then online dispute resolution. A network-policy summary lists a first-level expectation of 2 hours to respond and 24 hours to offer a resolution. An official IGM SDK exists for Node. Build: a copilot that ingests issue events, classifies them, proposes responses inside the 2-hour window, gathers evidence from order and logistics systems, and escalates to the grievance officer.

BRIEF D — Bring your own client

A real employer or design partner, under confidentiality rules. The same phases, hidden-scenario discipline and defence apply. The pod agrees scope, data access and success measures with the client during the 48-hour discovery window.

ACCEPTANCE

DEFINITION OF DONE	The deployed system, runbook and executive readout are delivered and match the agreed SOW.
HIDDEN SCENARIOS	The hidden scenario suite is passed.
GRADUATION	The 12 graduation questions are answered live.
TIME	8 live + 16 outside hours.

WEEK 11 · DESIGN

Capstone — Discover *& Design*

WEEK OBJECTIVE

Choose a brief, run a 48-hour discovery with a client role, and leave the week with a design you can be held to.

IDENTITY SHIFT *Owner → Engagement lead*

CASE 6 | PROJECT: CAPSTONE ENGAGEMENT | WEEK A OF 2

S1 Discover — Brief Selection, Client Interviews & Problem Brief

SATURDAY · 2 HRS

BUILDS ON	Weeks 1–10: six skills practised in blocks. Now you do the whole lifecycle on a fresh brief, with a pod.
TOPICS	• Select a capstone brief (A, B, C or bring your own client) • Run a 48-hour discovery window • Interview client roles and resolve contradictory asks • Write the problem brief and success measures • Agree scope and the SOW
CLAUDE IN USE	Claude as research partner across the public sources for your brief. You verify every fact.
BUILD	Kickoff and 48-hour discovery with a mentor-played client. Draft the problem brief and SOW.
TOOL STACK	Claude · public sources · docs

S2 Design — Architecture, Threat Model & First Working Slice

SUNDAY · 2 HRS

BUILDS ON	S1: the problem is agreed. Now you commit to an architecture and show a first slice working.
TOPICS	• Architecture with architecture decision records • Threat model and risk register • Evaluation plan and hidden-scenario plan • Data plan • First working slice
CLAUDE IN USE	Claude Code builds the first slice. You defend the architecture.
BUILD	Design review with a mentor as CTO. Deliver the architecture with ADRs, threat model, eval plan, data plan, risk register and a first working slice. Curveball: a requirement change plus a data-quality problem.
TOOL STACK	Claude Code · GitHub · everything from Weeks 1–10

WEEKLY ASSIGNMENT

TITLE	Capstone Design Pack
BRIEF	Deliver the problem brief, SOW, architecture with ADRs, threat model, eval plan, data plan, risk register and first working slice, and show how you handled the curveball.
DELIVERABLES	• Design pack • Repo • Half-page executive summary for the client
TIME	7–8 hrs
ASSESSED BY	Mentor review in your 15-minute 1:1, checked against the week's ten competencies.
FEEDS INTO	Week 12: you build, harden and defend this design.

PROJECT 6 MILESTONE · FOUNDATION CHECKPOINT

PROJECT	Capstone Engagement
DUE THIS WEEK	• Problem brief and SOW • Architecture with ADRs, threat model and risk register • Eval plan, data plan and hidden-scenario plan • First working slice

WEEK OUTCOME

A design the client agreed to and the pod can be graded against.

WEEK 12 · DELIVER

Capstone — Build, Harden & *Demo Day*

WEEK OBJECTIVE

Build, harden and deliver the capstone, then defend it live on Demo Day against a change request you have not seen.

IDENTITY SHIFT *Engagement lead → AI FDE*

CASE 6 | PROJECT: CAPSTONE ENGAGEMENT | WEEK B OF 2

S1 Build & Harden — Hidden Scenarios, Evals & Security

SATURDAY · 2 HRS

BUILDS ON	Week 11: an agreed design and a first slice. Now it has to survive scenarios you have not seen.
TOPICS	• Build against the hidden scenario suite • Run the eval harness and fix the top failures • Security review against the threat model • Deploy with tracing and cost controls • Write the runbook and executive summary
CLAUDE IN USE	Claude Code accelerates the build. You decide what ships.
BUILD	Build and harden clinic with mentors acting as client, security reviewer and CTO.
TOOL STACK	Claude Code · eval harness · OpenTelemetry · CI

S2 Demo Day — Demo, Live Defence & Executive Readout

SUNDAY · 2 HRS

BUILDS ON	S1: the system is hardened and deployed. Now you show it and answer for it.
TOPICS	• The 15-minute demo • The 15-minute live defence with an instructor-injected change request • Executive readout to an industry panel • The 12 graduation questions, answered live • The next-build roadmap
CLAUDE IN USE	Claude as rehearsal partner for the panel questions. On stage, you answer alone.
BUILD	Demo Day: 15-minute demo, 15-minute live defence with a change request, and an executive readout to an industry panel.
TOOL STACK	Your deployed system · your runbook

WEEKLY ASSIGNMENT

TITLE	Capstone Delivery — Ship, Show, Defend
BRIEF	Deliver the deployed capstone, the runbook and the executive readout. Acceptance: the Definition of Done is met, the hidden scenario suite is passed and the 12 graduation questions are answered live.
DELIVERABLES	• Deployed system • Runbook • Readout; live Demo Day • Half-page executive summary for the client
TIME	7–8 hrs
ASSESSED BY	Mentor review in your 15-minute 1:1, checked against the week's ten competencies.
FEEDS INTO	Graduation: the portfolio and the Scenario Library you keep.

PROJECT 6 MILESTONE · FINISHED PROJECT

PROJECT	Capstone Engagement
DUE THIS WEEK	• Deployed system with tracing and cost controls • Runbook and executive readout • Hidden scenario suite passed • Demo Day and 12 graduation questions answered live
GATE	Demo Day

COMPONENT OWNER	The component you own and defend alone: _____
POD ROLES	Client lead · Quality lead · Delivery lead (rotates each case)
MENTOR SIGN-OFF	Rubric level: Not yet / Meets / Strong / Hire-ready Date: _____
PORTFOLIO	Repository, report and defence record added to your public portfolio.

WEEK OUTCOME

A defended engagement from ambiguous brief to deployed system.

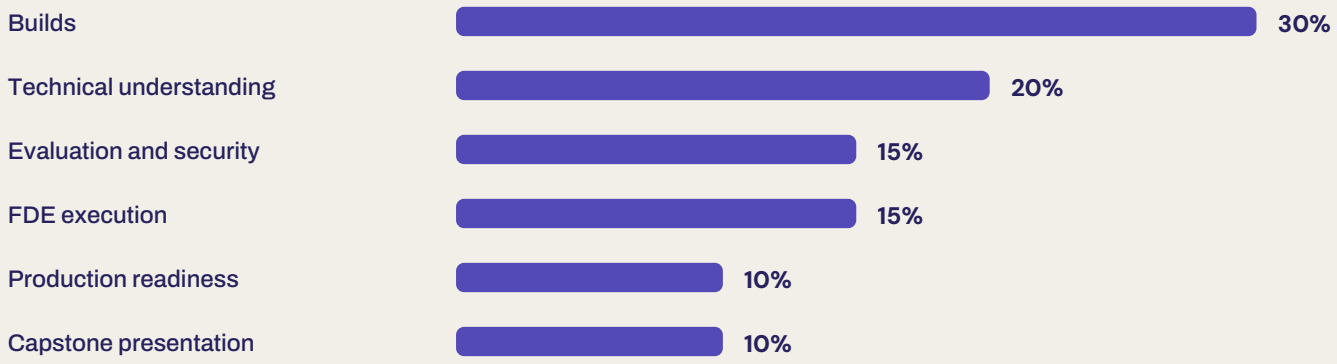
THE STANDARD

Assessment & *Grading*

Graded by defence: show it, explain it, extend it live.

You clear a competency by showing an artefact you built or by explaining and extending it live. You need 80 of 100 competencies and a capstone pass to graduate.

SCORE WEIGHTS



FOUR-LEVEL RUBRIC

NOT YET	The work does not meet the acceptance criteria, or you cannot explain it. Resubmit and defend again.
MEETS	The criteria are met and you can explain your component. A case started from a checkpoint repo is capped here.
STRONG	The criteria are met with evidence, and you handle the live change request and hidden scenarios with sound reasoning.
HIRE-READY	You could run this engagement with a client tomorrow: the numbers are defensible and the trade-offs are your own.

THE GATES

CASE DEFENCES	End of every two-week block, Cases 1 to 5.
MID-POINT DEFENCE	Closes Case 3, at Week 6.
WAR ROOM	Case 4: pods attack each other's hardened assistants.
INCIDENT DRILL	Case 5: a live provider outage plus a cost spike.
DEMO DAY	Capstone: demo, live defence and executive readout.

THE 12 GRADUATION QUESTIONS

01 What problem are you solving, and why does it need AI?

02 Why this model and this architecture?

03 Why RAG and not fine-tuning?

04	Why an agent and not a plain workflow?
05	How did you evaluate it, and how reliable is that number?
06	What happens when it fails?
07	How did you secure it?
08	What does it cost per request?
09	How do you trace and monitor it?
10	How would you hand it over?
11	What does the client do on day 30?
12	What would you build next?

YOUR PORTFOLIO

PROJECTS	Six milestone projects, each with its repository, report and defence record
WEEKLY WORK	Twelve weekly assignments
PROOF	A public MCP server, a go / no-go certification memo, a pilot postmortem, an ROI model and a written case study
SCENARIO LIBRARY	Your personal library of hidden-scenario tests

THE REGISTER

The 100 *Competencies*

Ten a week, five per session, each cleared by an artefact or an oral defence.

WEEK 01 — Discover, Frame & Model Basics

S1

- C01 FDE operating model and lifecycle
- C02 Discovery interviews and process mapping
- C03 Use-case qualification (value, feasibility, risk, kill criteria)
- C04 AI vs ML vs rules
- C05 Success metrics and baselines

S2

- C06 LLM mechanics (tokens, context, sampling)
- C07 How models are trained and why they hallucinate
- C08 Model landscape and selection
- C09 Stakeholder mapping and contradictory asks
- C10 Claude as research and synthesis partner

WEEK 02 — Model Behaviour, Prompting & Context

S1

- C11 System prompts and persona design
- C12 Structured outputs and schema enforcement
- C13 Reasoning models and thinking budgets
- C14 Reasoning patterns (CoT, ReAct, Reflexion)
- C15 Context budgeting and compaction

S2

- C16 Prompt caching and batching
- C17 Grounding and citations
- C18 Prompt versioning and regression
- C19 Multilingual and Indic model choices
- C20 LLM vs classical-baseline benchmark

WEEK 03 — Build with Claude Code & Data Foundations

S1

- C21 Claude Code workflow (CLAUDE.md, plan, implement, test)
- C22 Extending Claude Code (skills, subagents, hooks, plugins)
- C23 Full-stack app (UI, API, auth)
- C24 Git and diff review of AI-written code
- C25 Testing (unit, integration, test-first with AI)

S2

- C26 SQL, schema design, data modelling
- C27 Data quality and ingestion
- C28 PII classification and DPDP basics
- C29 AI UX (confidence, fallbacks, feedback)
- C30 PRD-lite and SOW-lite

WEEK 04 — Knowledge & Retrieval

S1

- C31 Parsing and ingestion (PDF, OCR, tables)
- C32 Chunking
- C33 Embeddings and model choice
- C34 Vector stores and ANN
- C35 Freshness, dedup and conflicting sources

S2

- C36 Hybrid search and reranking
- C37 Citations and grounded answers
- C38 Retrieval metrics and failure analysis
- C39 Query rewriting and agentic RAG
- C40 Knowledge graphs

WEEK 05 — Agent Architecture & Memory

S1

- C41 Tool design and schemas
- C42 Agent loop and bounded autonomy
- C43 Workflow vs agent decision
- C44 Human-in-the-loop approvals
- C45 Failure handling

S2

- C46 Building an MCP server (2026-07-28 stateless spec)
- C47 MCP authorization and tool permissions
- C48 Short-term and long-term memory
- C49 Durable workflows
- C50 Computer-use and browser agents for systems without APIs

WEEK 06 Multi-Agent & Enterprise Integration

S1

- C51 Multi-agent patterns (router, supervisor, parallel, handoff)
- C52 Subagents and the Claude Agent SDK
- C53 Shared state and communication
- C54 Channels (WhatsApp, email)
- C55 Integration testing with sandboxes

S2

- C56 REST/GraphQL, webhooks, idempotency
- C57 OAuth2 and connectors
- C58 Workflow orchestration and queues (n8n)
- C59 Legacy systems (SOAP/XML, SQL Server)
- C60 Rate limits, backoff, dead-letter handling

WEEK 07 Evaluation & Quality

S1

- C61 Eval design and failure taxonomy
- C62 Golden sets with client SMEs
- C63 LLM-as-judge (rubric, pairwise, calibration)
- C64 RAG evals
- C65 Agent trajectory and tool-use evals

S2

- C66 Eval statistics (sample size, variance, annotator agreement)
- C67 CI regression gates
- C68 Online evals, feedback and drift
- C69 Eval platforms and experiment tracking
- C70 Reporting quality to a client

WEEK 08 Safety, Security & Governance

S1

- C71 Threat modelling (OWASP LLM Top 10)
- C72 Prompt injection, direct and indirect
- C73 Red-team method and war room
- C74 Tool abuse, least privilege, blast radius
- C75 Bias, fairness, abuse testing

S2

- C76 Guardrails and PII redaction
- C77 Identity and permissions (RBAC, permission-aware retrieval)
- C78 Data residency, retention, DPDP and sector rules
- C79 Audit trails, lineage, explainability
- C80 Risk review and security questionnaire

WEEK 09 Deploy & Operate

S1

- C81 Environments, secrets, config
- C82 Docker
- C83 CI/CD
- C84 Cloud deployment with IAM and networking basics
- C85 Release safety (canary, A/B, shadow, rollback)

S2

- C86 Observability (logs, traces, token and latency metrics)
- C87 SLOs, error budgets, alerting
- C88 Cost control (caching, batching, routing, budget guards)
- C89 Incident response and postmortems
- C90 UAT plan, runbook, operational handoff

WEEK 10 Economics, Delivery & Ownership

S1

- C91 Model selection matrix and routing
- C92 Unit economics per request
- C93 ROI proof and business case
- C94 Vendor and model risk, and the fine-tune vs RAG vs prompt decision
- C95 Adoption metrics and change management

S2

- C96 SOW, commercials, pricing negotiation
- C97 Executive readout
- C98 Handover and enablement
- C99 Account expansion
- C100 Legal, IP and licensing awareness

WEEK 11 Capstone — Discover & Design

S1

- C100 Select a capstone brief (A, B, C or bring your own client)
- C100 Run a 48-hour discovery window
- C100 Interview client roles and resolve contradictory asks
- C100 Write the problem brief and success measures
- C100 Agree scope and the SOW

S2

- C100 Architecture with architecture decision records
- C100 Threat model and risk register
- C100 Evaluation plan and hidden-scenario plan
- C100 Data plan
- C100 First working slice

WEEK 12 Capstone — Build, Harden & Demo Day

S1

- C100 Build against the hidden scenario suite
- C100 Run the eval harness and fix the top failures
- C100 Security review against the threat model
- C100 Deploy with tracing and cost controls
- C100 Write the runbook and executive summary

S2

- C100 The 15-minute demo
- C100 The 15-minute live defence with an instructor-injected change request
- C100 Executive readout to an industry panel
- C100 The 12 graduation questions, answered live
- C100 The next-build roadmap

THE TOOLKIT

The Complete *Toolkit*

Every tool used across the twelve weeks, grouped by the job it does.

CLAUDE PLATFORM

Claude API	Structured output, tool use, caching and batching. The model behind every build.	Wk 2–12
Claude Code	Directs plan, implement, test and review; extended with skills, subagents, hooks and plugins.	Wk 1–12
Claude Agent SDK	Builds agents and subagents with typed tools and bounded autonomy.	Wk 5–6
MCP	Publishes your tools as a permissioned server any compatible client can use (2026-07-28 spec).	Wk 5–6

DATA & RETRIEVAL

Postgres + pgvector	Relational schema, PII tagging and vector search in one place.	Wk 3–4
Qdrant	A dedicated vector store for comparison against pgvector in the retrieval bake-off.	Wk 4
Neo4j	Optional graph path for linked knowledge in the farm advisory case.	Wk 4
OCR tooling	Turns scanned advisories into text, with sample checks on accuracy.	Wk 4

ORCHESTRATION & INTEGRATION

n8n	Workflow automation for queues, webhooks and supplier follow-ups.	Wk 6
REST, webhooks, OAuth2	The connective tissue to enterprise systems, with idempotency and retries.	Wk 6, 9
WhatsApp / email sandboxes	Safe channels for outreach agents before anything reaches a real customer.	Wk 6, 9

EVALUATION & SAFETY

promptfoo · Braintrust	Regression suites, experiment tracking and CI gates.	Wk 2, 7–8
RAGAS	Metrics for retrieval and grounded answers.	Wk 7
Presidio	Detects and redacts personal data before it reaches logs.	Wk 8
NeMo Guardrails	Policy rails on inputs, outputs and tool calls.	Wk 8

DELIVERY & OPERATIONS

Docker · GitHub Actions	Containers and pipelines with eval and security gates before release.	Wk 9
OpenTelemetry	Traces, token counts and latency across the whole system.	Wk 9, 12
Phoenix · Langfuse	Trace review and cost monitoring for LLM calls.	Wk 9
Cloud host	IAM, networking basics and a deployed environment with a rollback.	Wk 3, 9

COLLABORATION

GitHub	Repos, pull requests and diff review of AI-written code.	Wk 1–12
Docs and spreadsheets	Memos, SOWs, ROI models and executive summaries.	Wk 1–12

THE GUARDRAILS

Data, Ethics & *Ground Rules*

How we keep every case safe, honest and independent.

SYNTHETIC DATA	All case data is synthetic. No real taxpayer, patient, farmer, customer or citizen data is used in any case.
INDEPENDENT BRIEFS	Briefs are Menler-authored from public information. No engagement with, or endorsement by, any named body is implied.
RECOMMEND, NEVER DECIDE	Systems in the health, tax and claims cases recommend and explain. They never deny claims, file returns or decide eligibility. A human decides.
MESSAGING CONSENT	Case 5 needs WhatsApp opt-in, template approval, immediate opt-out handling and a consent notice for any live pilot.
LICENSING	The court-judgments capstone brief requires a licence check on the corpus before any data is used.
TIME-SENSITIVE RULES	GST rules change often and a chartered accountant reviews Case 3 before each cohort. DPDP obligations phase in to about May 2027.
VERIFIED SOURCES	Statistics from vendors or secondary sources are labelled as claims, and you are taught to measure them, not trust them.

WHAT WE EXPECT FROM YOU

OWNERSHIP	Own your component and be able to defend every line of it.
HONESTY	Say "I don't know" in the system and in the room when that is the honest answer.
DATA CARE	Never paste real client or personal data into a prompt.
DISCLOSURE	Disclose where Claude wrote code you did not review.
CLIENT RESPECT	Treat mentor-played clients as real clients.

Case Sources *& Notes*

Public sources behind the case briefs. Figures are re-verified before every cohort.

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Note: one secondary source reports that a revised RBI Integrated Ombudsman Scheme may supersede the 2021 scheme. Menler checks the RBI site before each cohort and updates Case 1 accordingly.

THE OUTCOME

What You *Leave With*

Evidence, judgment and a cohort.

A shipped engagement	Six milestone projects, each defended to a panel, and a capstone taken from ambiguous brief to deployed system.
Evidence you can show	A public MCP server, a go / no-go memo, a pilot postmortem, an ROI model and a written case study.
A Scenario Library	Roughly sixty hidden-scenario tests from the cases, to reuse on your own future builds.
A working vocabulary	The language of baselines, confidence intervals, SLOs, approvals and handover, spoken to clients and CFOs.
Defended judgment	Answers to twelve hard questions, given live, about a system you designed.
A cohort	Pod-mates, mentors and a network of people who have shipped the same engagements.

THE PROMISE

You will leave not just knowing how AI systems work, but having been held accountable for one, from the first client interview to the last defence.